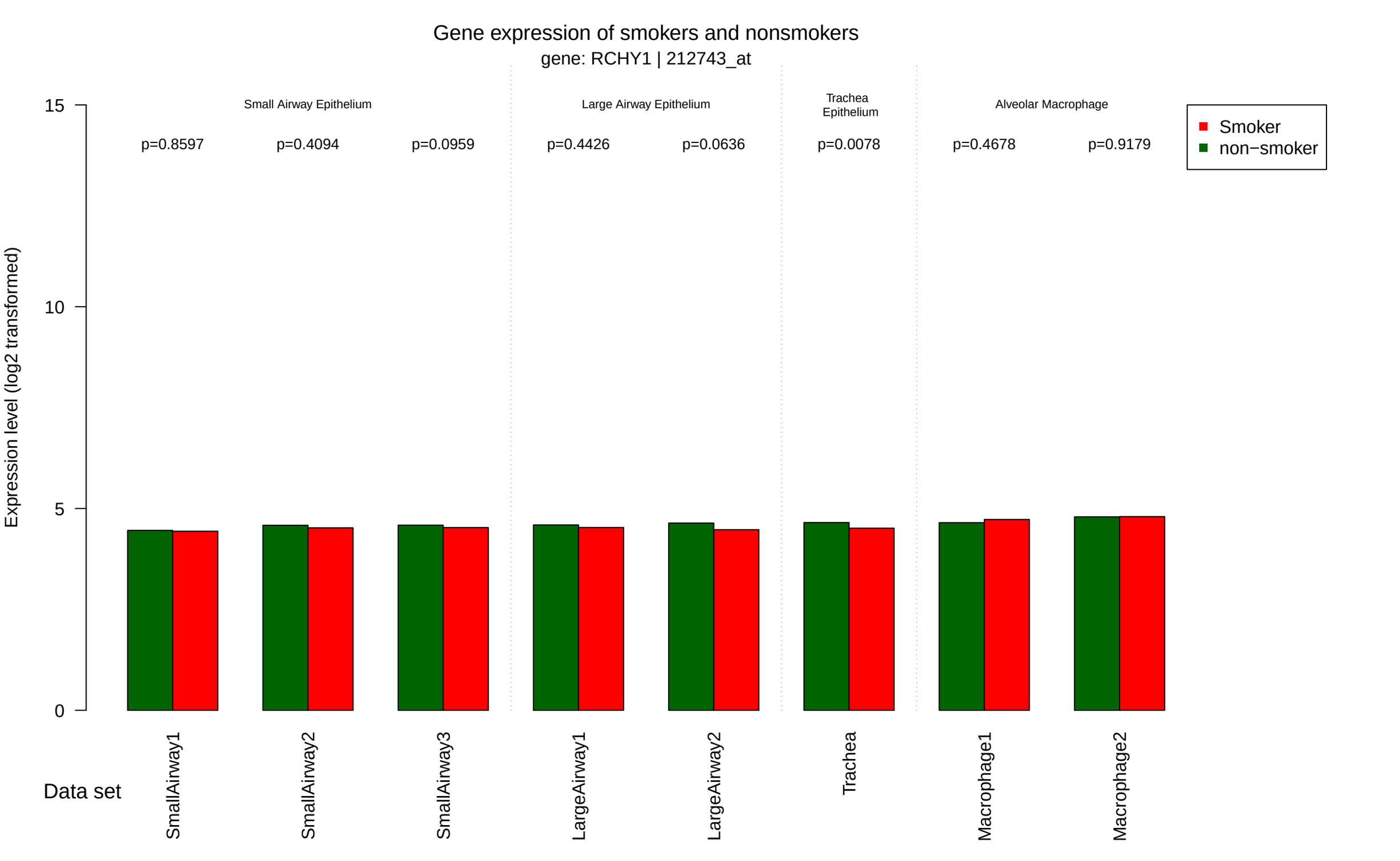
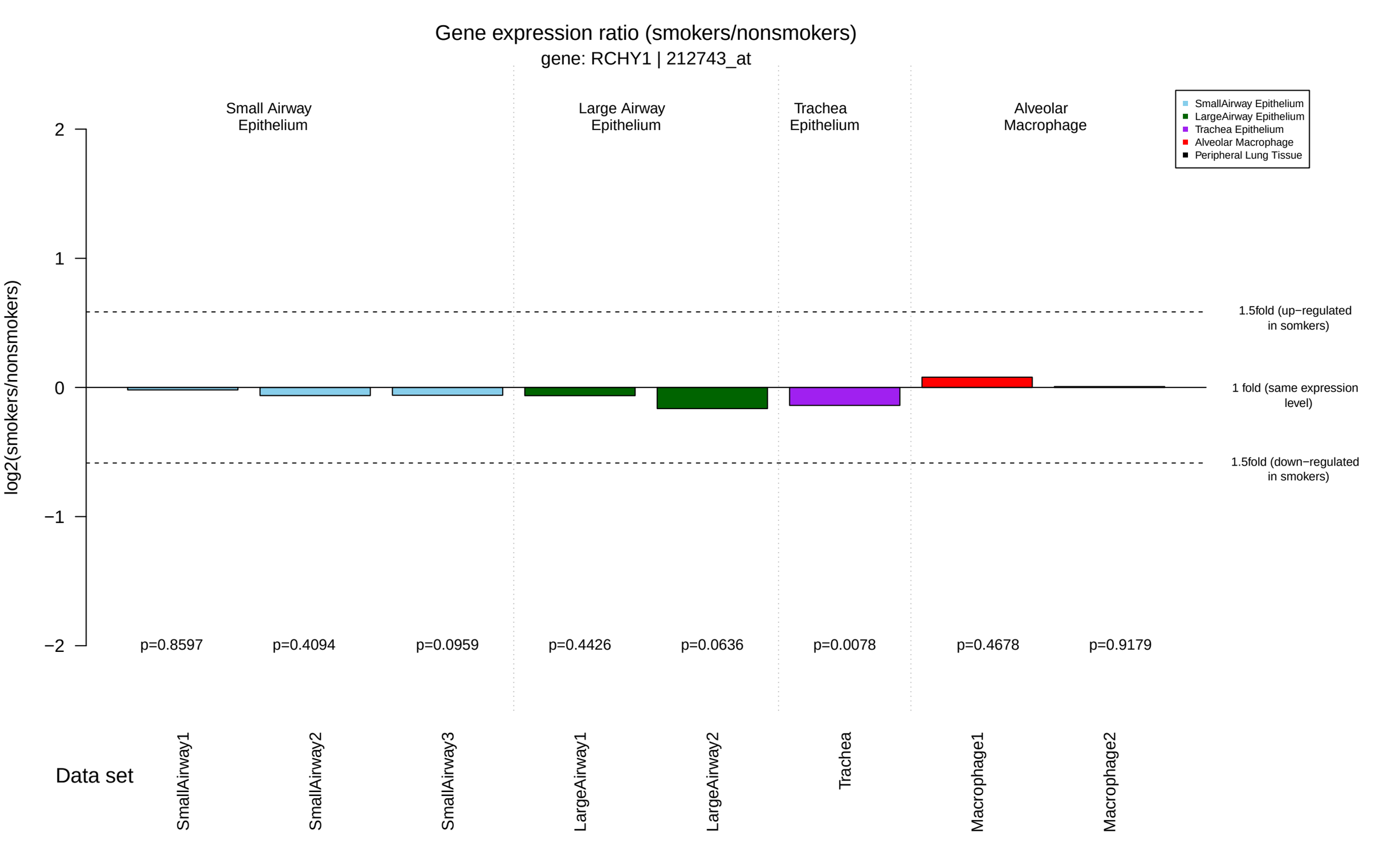
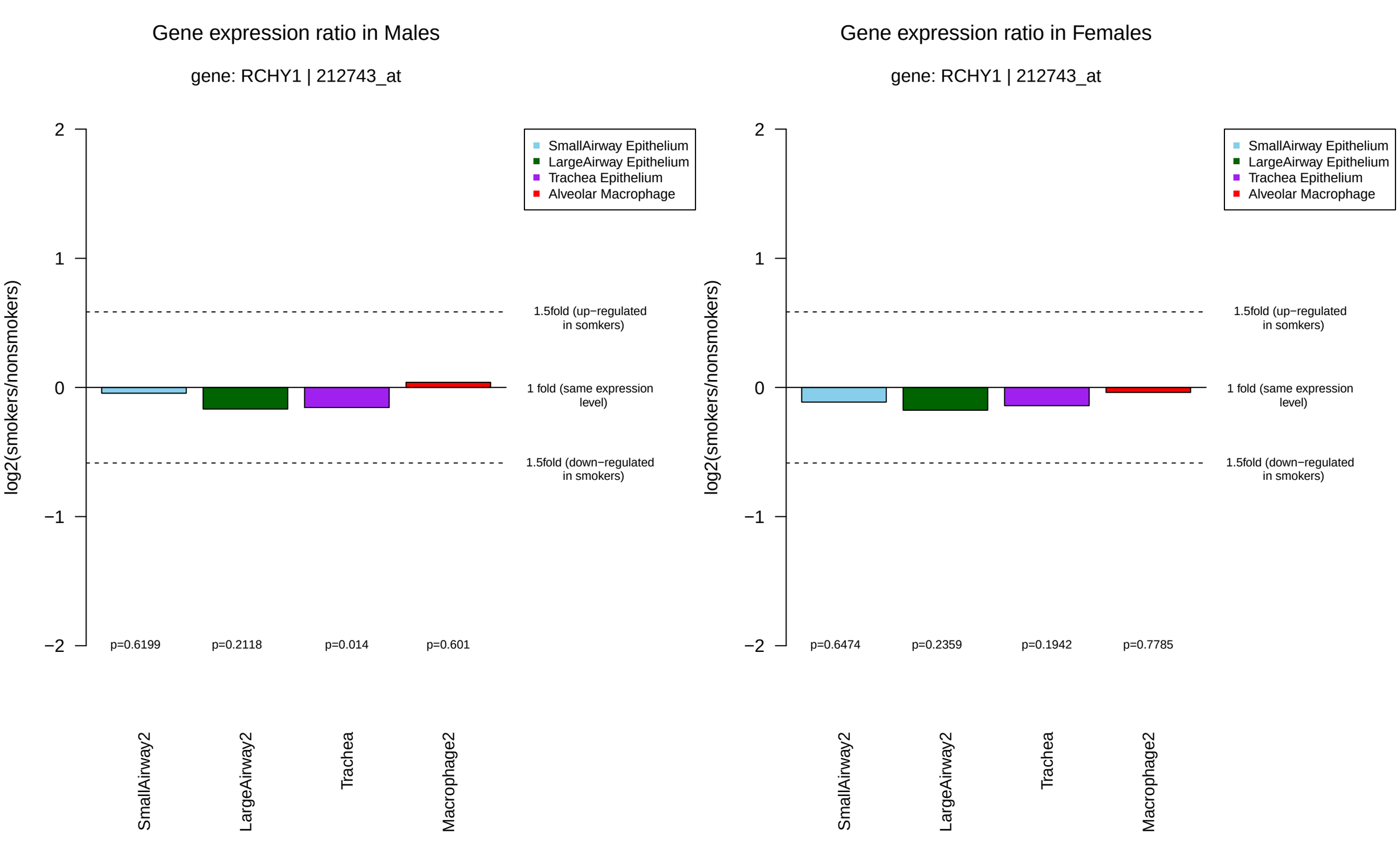
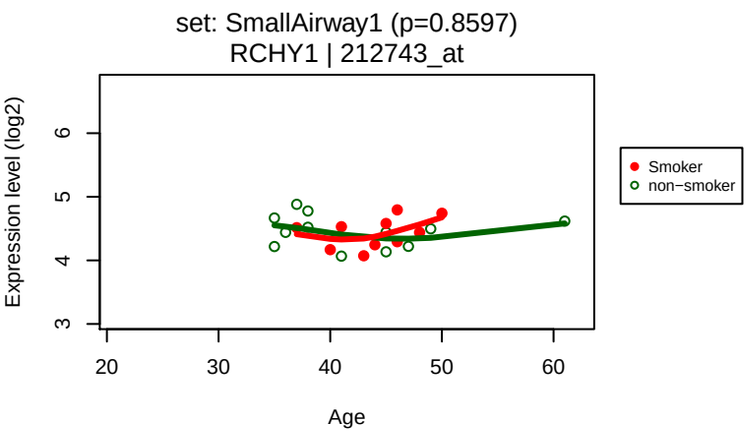
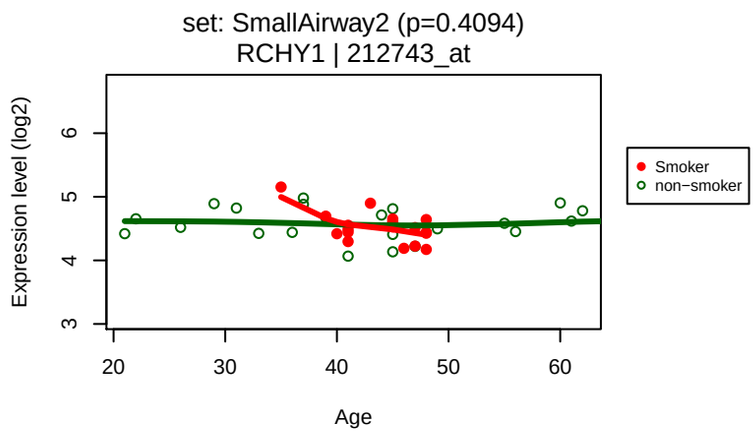
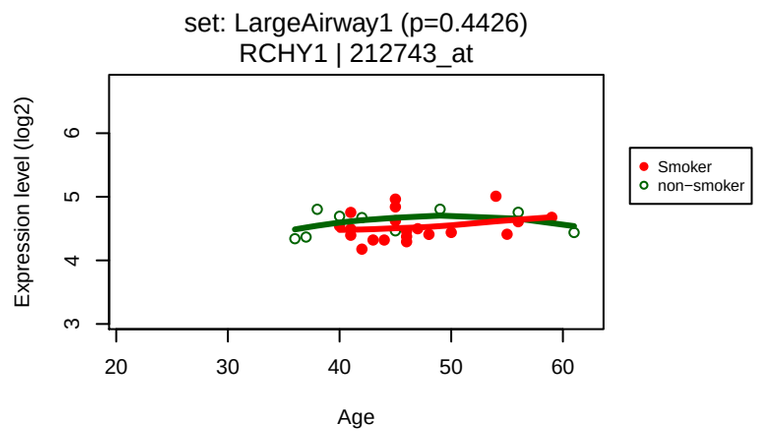

Gene expression of smokers and nonsmokers
gene: RCHY1 | 212743_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8597
p=0.4094
p=0.0959
p=0.4426
p=0.0636
p=0.0078
p=0.4678
p=0.9179
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

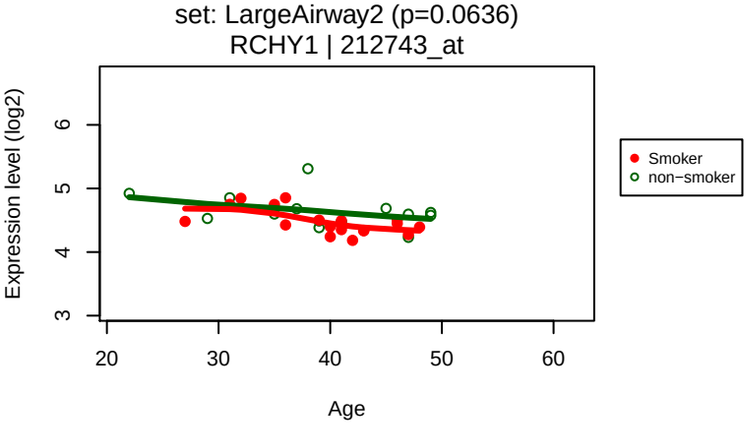

Gene expression ratio (smokers/nonsmokers)
gene: RCHY1 | 212743_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8597
p=0.4094
p=0.0959
p=0.4426
p=0.0636
p=0.0078
p=0.4678
p=0.9179
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

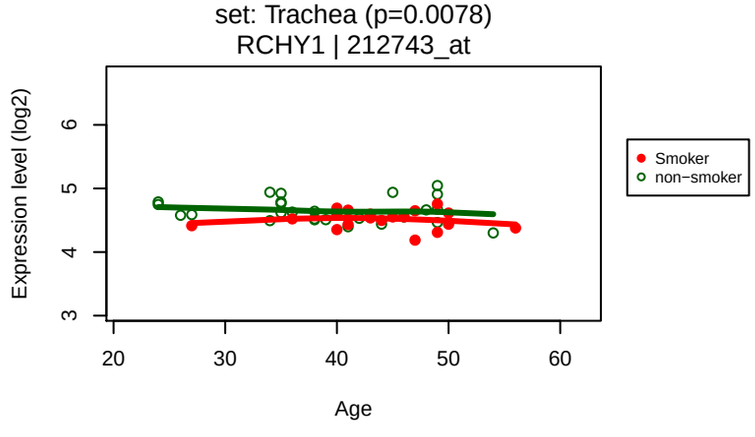

Gene expression ratio in Males
gene: RCHY1 | 212743_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6199
p=0.2118
p=0.014
p=0.601
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCHY1 | 212743_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6474
p=0.2359
p=0.1942
p=0.7785
SmallAirway2
LargeAirway2
Trachea
Macrophage2

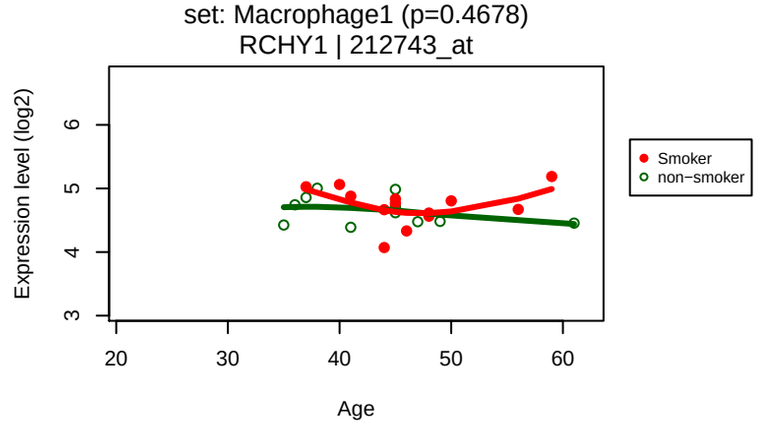

set: SmallAirway1 (p=0.8597)
RCHY1 | 212743_at
Expression level (log2)
Age
Smoker
non−smoker

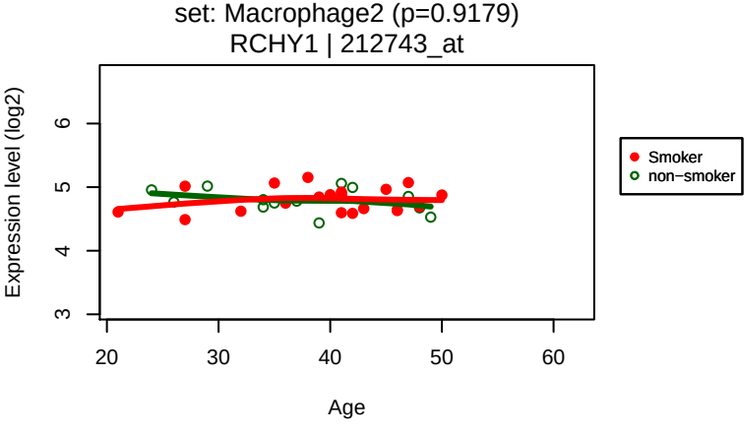

set: SmallAirway2 (p=0.4094)
RCHY1 | 212743_at
Expression level (log2)
Age
Smoker
non−smoker

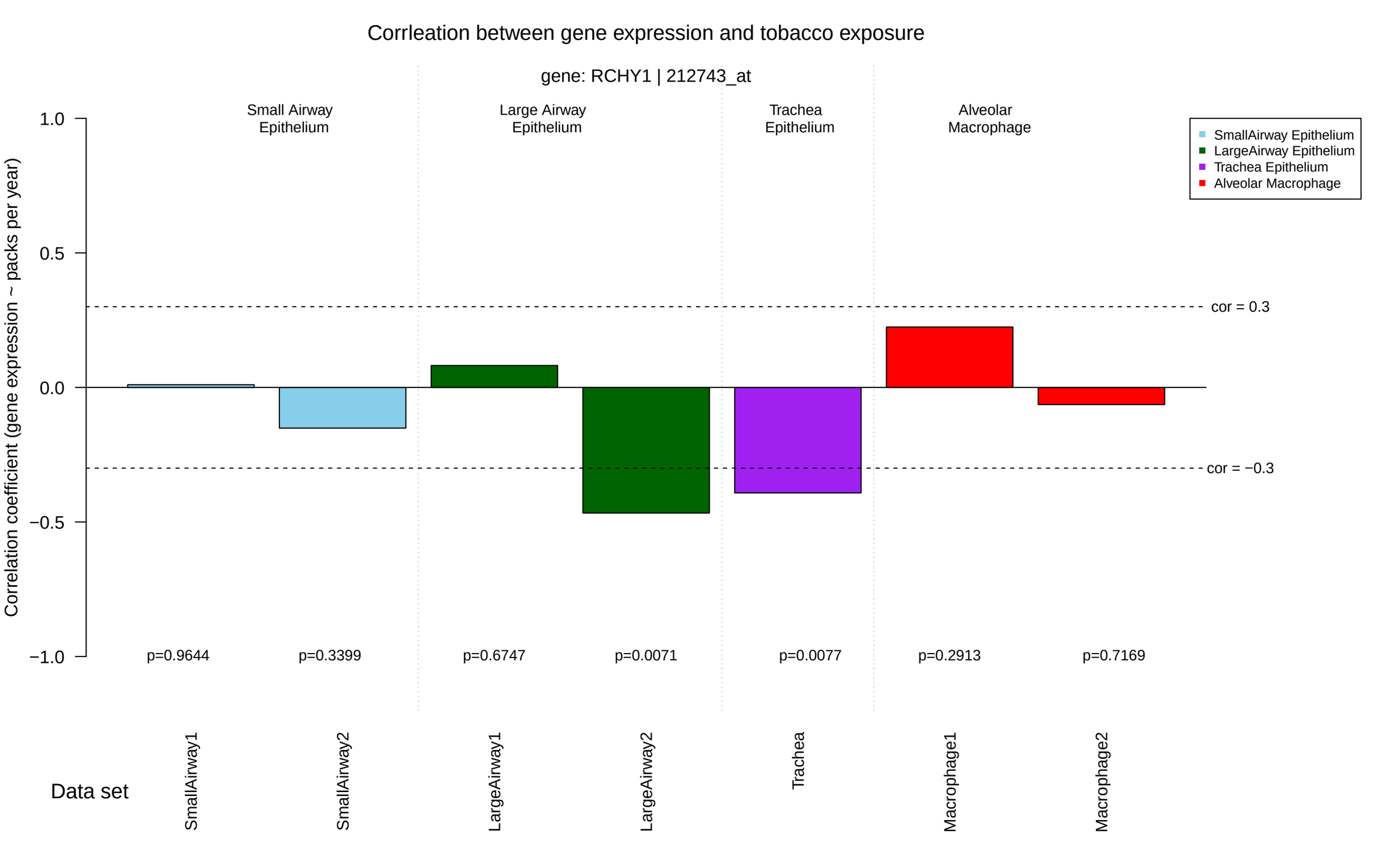

set: LargeAirway1 (p=0.4426)
RCHY1 | 212743_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0636)
RCHY1 | 212743_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0078)
RCHY1 | 212743_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4678)
RCHY1 | 212743_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9179)
RCHY1 | 212743_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCHY1 | 212743_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9644
p=0.3399
p=0.6747
p=0.0071
p=0.0077
p=0.2913
p=0.7169
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2